Project Design Phase-II
Technology Stack (Architecture & Stack)

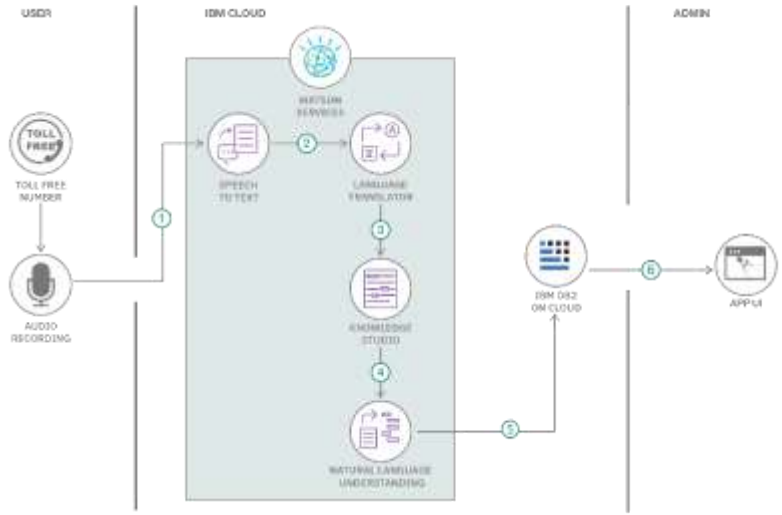
Date	3.10.2026
Team ID	6ab8c3ef30f9767f87730407
Project Name	Streamlining IT Procurement: Automating Standard Laptop order with flow designer
Maximum Marks	4 Marks

Technical Architecture:

The deliverable includes the project architecture and the information as per Table-1 and Table-2.

Architectural Diagram: see Figure 1 on the next page.

Architecture Flow: Service Catalog (Standard Laptop) → Approval Condition → Flow Assignment → Standard Laptop Task Flow (Flow Designer) → Catalog Task under the related Requested Item → Hardware assignment group.https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/



Guidelines:

- Include all the processes (As an application logic / Technology Block)
- Provide infrastructural demarcation (Local / Cloud)
- Indicate external interfaces (third party API's etc.)
- Indicate Data Storage components / services
- Indicate interface to machine learning models (if applicable)

Figure 1: Technical Architecture – Standard Laptop Order Automation

Service Catalog (Standard Laptop) → Approval Condition → Flow Assignment → Standard Laptop Task Flow → Catalog Task → Hardware assignment group

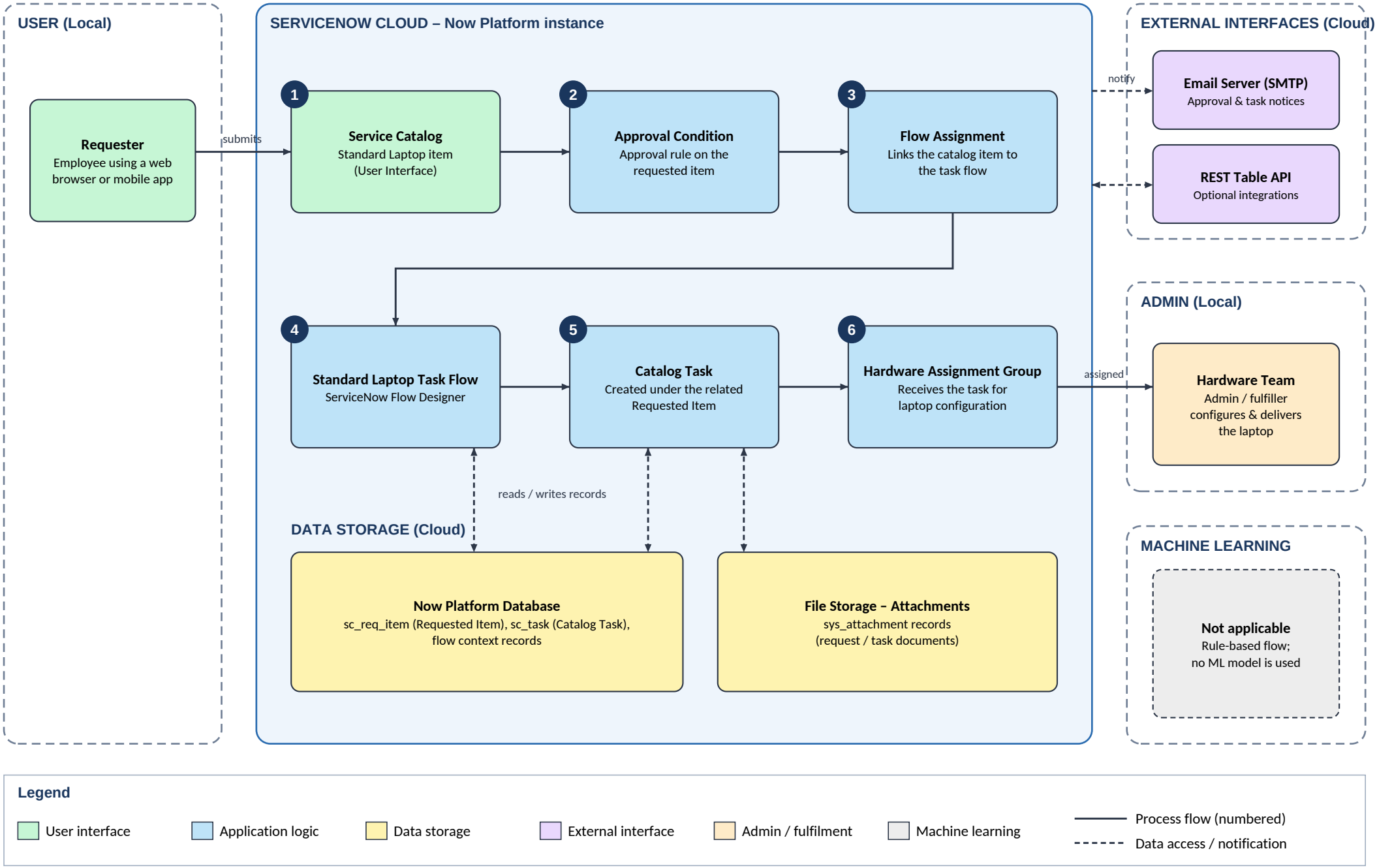


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	Users submit laptop requests through the Standard Laptop Service Catalog item.	ServiceNow Service Catalog
2	Application Logic-1	Automates the creation of a Catalog Task for Standard Laptop requests once the configured approval condition is reached.	ServiceNow Flow Designer – Standard Laptop Task Flow
3	Application Logic-2	Connects the Standard Laptop Service Catalog item with the Standard Laptop Task Flow Designer flow so the required automation is executed.	ServiceNow Flow Assignment configuration
4	Application Logic-3	Creates a Catalog Task under the related Requested Item and assigns it to the Hardware assignment group for laptop configuration.	ServiceNow Catalog Task / Flow Designer
5	Database	Stores and relates the Requested Item and Catalog Task records used by the workflow.	ServiceNow platform data records
6	Cloud Database	Cloud-hosted instance database that persistently holds the request, approval and task records.	ServiceNow Now Platform cloud database
7	File Storage	Stores attachments added to the request or catalog task, such as approval or configuration documents.	ServiceNow Attachments (sys_attachment)
8	External API-1	Emails notify requesters, approvers and the Hardware group of approval and task updates.	Email server (SMTP) / ServiceNow Notifications
9	External API-2	Optional interface that lets other systems read or create requests and tasks.	ServiceNow REST Table API
10	Machine Learning Model	Not applicable – the workflow is condition-based and does not use a machine learning model.	None (rule-based Flow Designer logic)
11	Infrastructure (Server / Cloud)	Platform on which the Service Catalog and Flow Designer automation are executed.	ServiceNow Now Platform (cloud-hosted instance)

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	The Service Portal catalog front end is built on open-source web frameworks; custom logic uses JavaScript.	AngularJS, Bootstrap, JavaScript
2	Security Implementations	Role-based access and ACLs control who can request, approve and fulfil tasks; data is encrypted in transit.	ServiceNow Roles & ACLs, HTTPS/TLS
3	Scalable Architecture	The automation reduces manual task creation and routes approved Standard Laptop requests to the appropriate team.	ServiceNow Flow Designer
4	Availability	Runs on ServiceNow's cloud, which provides redundant infrastructure so the catalog and flow remain accessible.	ServiceNow Cloud (redundant data centres)
5	Performance	The automated workflow is intended to reduce manual task creation, improve processing speed, and minimize user wait times.	ServiceNow Flow Designer

References:

https://c4model.com/

https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/

https://www.ibm.com/cloud/architecture

https://aws.amazon.com/architecture

https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d